

MOUNT SHASTA, CA, USA

WMO: 725957

Lat: 41.333N

Lon: 122.333W

Elev: 3535

StdP: 12.91

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24215

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	18.2	21.7	6.7	8.9	29.1	11.5	11.3	30.3	13.4	38.6	12.2	38.0	1.7	50	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.0	91.9	64.2	89.4	63.1	86.4	61.7	67.7	86.1	65.6	84.8	63.8	82.9	2.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	91.5	78.2	57.3	79.9	76.0	55.1	73.7	74.0	33.9	86.0	32.1	84.7	30.7	82.8	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.5	9.1	DB	10.7	97.2	5.1	2.7	7.1	99.1	4.1	100.7	1.2	102.2	-2.5	104.2	(4)
(5)			WB	9.5	70.7	4.9	3.0	6.0	72.9	3.1	74.7	0.3	76.4	-3.3	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	36.7	38.4	42.4	46.9	55.1	62.4	69.4	67.6	61.7	51.2	41.5	35.6		
	DBStd	13.34	5.83	5.89	6.54	7.27	7.43	7.02	5.05	4.74	6.34	6.47	6.41	5.91		
	HDD50	1948	414	326	249	144	36	5	0	0	3	63	262	446		
	HDD65	5529	879	745	702	542	316	128	17	26	131	428	704	911		
	CDD50	2246	0	1	13	52	193	375	602	547	354	101	8	0		
	CDD65	350	0	0	0	0	7	48	154	108	32	1	0	0		
	CDH74	7196	0	0	1	43	314	964	2641	2165	987	81	0	0		
(13)	CDH80	2761	0	0	0	3	76	311	1156	879	323	13	0	0		
(14)	Wind	WSAvg	3.1	3.2	3.3	3.6	3.9	3.6	2.4	2.2	2.6	2.9	2.9	3.3		
(15) Precipitation	PrecAvg	40.6	7.0	6.0	5.3	2.9	1.9	1.0	0.3	0.4	0.7	2.2	5.6	7.0		
	PrecMax	75.2	27.5	21.8	18.9	9.1	9.3	3.8	1.7	2.6	3.9	9.7	17.2	25.9		
	PrecMin	14.2	0.2	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.1		
	PrecStd	13.0	5.3	4.9	4.3	2.2	2.0	1.0	0.4	0.6	0.9	2.2	4.2	5.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	64.3	71.0	79.2	87.0	92.0	96.8	95.2	91.1	81.4	67.6	54.9		
		MCWB	44.6	48.2	51.6	55.4	60.2	64.6	66.4	64.3	61.7	58.7	51.9	44.5		
	2%	DB	54.0	58.7	65.8	73.5	81.7	87.3	92.8	90.9	87.3	75.3	62.7	50.6		
		MCWB	42.7	45.9	49.1	53.6	59.6	62.9	65.1	63.6	59.8	55.4	49.7	42.9		
	5%	DB	49.6	53.6	61.4	68.8	77.3	83.5	90.0	88.2	83.8	71.8	57.2	47.8		
		MCWB	41.8	43.1	46.8	51.3	56.6	61.3	63.8	62.3	59.0	53.5	46.6	42.3		
	10%	DB	46.0	49.0	56.0	63.3	72.7	80.0	86.8	85.2	80.5	66.6	53.1	45.2		
		MCWB	40.7	41.6	44.8	49.3	54.9	59.9	63.0	60.9	58.4	51.4	44.6	41.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	50.1	53.8	57.7	64.1	68.8	71.1	69.5	66.0	59.8	54.2	48.0		
		MCDB	53.0	61.0	67.8	74.5	81.4	85.7	89.2	86.4	85.1	76.2	64.0	51.0		
	2%	WB	45.3	47.4	50.5	54.7	60.6	65.5	68.3	66.3	62.6	57.2	51.0	45.5		
		MCDB	50.3	55.5	62.5	70.9	78.5	83.7	86.6	85.6	81.5	73.0	59.6	48.3		
	5%	WB	43.3	44.8	48.2	52.2	58.3	62.9	66.2	64.2	60.4	54.7	48.3	43.7		
		MCDB	48.2	51.5	58.9	66.5	75.0	80.6	85.5	83.8	79.8	68.9	55.3	46.8		
	10%	WB	41.2	42.4	45.8	49.7	55.7	60.7	64.3	62.4	58.7	52.3	46.0	41.2		
		MCDB	45.6	48.3	54.6	61.5	70.7	77.9	83.7	82.3	78.3	64.7	51.2	44.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	17.1	19.9	22.8	25.9	28.3	33.0	34.0	32.1	26.0	18.5	14.5		
		MCDBR	23.1	27.0	31.0	34.7	35.0	34.5	37.0	37.5	38.7	35.1	27.2	20.2		
		MCWBR	14.5	16.2	17.2	18.0	16.5	15.9	15.9	16.2	17.5	18.2	16.8	13.5		
	5% WB	MCDBR	17.7	22.5	27.4	31.6	32.2	32.3	33.8	34.6	35.1	31.1	23.9	15.2		
		MCWBR	12.0	14.4	16.0	17.2	16.0	16.8	16.3	16.6	17.1	16.9	15.4	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.284	0.289	0.305	0.315	0.308	0.310	0.306	0.293	0.291	0.276	0.281		
	taud		2.553	2.563	2.579	2.529	2.536	2.590	2.596	2.590	2.611	2.618	2.623	2.550		
	Ebn at Noon		283	298	308	307	304	305	304	303	302	292	284	272		
	Edh at Noon		22	25	28	32	32	31	30	30	27	24	20	21		
(44) All-Sky Solar Radiation	RadAvg		620	909	1281	1753	2179	2521	2569	2279	1808	1204	712	513		
	RadStd		101	114	142	174	167	140	92	91	81	112	82	101		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.74	N/A	N/A	+2.18	N/A	N/A	N/A	N/A	+230	+86
(47)	Regional (15 neighbors)	N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	+169	+27

Nomenclature: See separate page